

Olimex launches an open hardware development board



Bulgarian open hardware company Olimex has launched its ESP32-P4-DevKit, a compact development board featuring Espressif's dual-core 400MHz RISC-V ESP32-P4 system-on-chip.

The kit is designed to be breadboard-friendly and comes in a compact, gumstick form factor. It includes a full-size Ethernet port, MIPI Camera Serial Interface (CSI), and Display Serial Interface (DSI) to take advantage of the onboard H.264 hardware codec. Additionally, 0.1" pin headers expose general-purpose input/output (GPIO) connectivity.

The board has Espressif's ESP32-P4, which integrates two 32-bit RISC-V cores running up to 400MHz, 768kB of static RAM, and support for up to 16MB of external SPI flash memory. A microSD card slot provides additional program and data storage. Other hardware features include SPI, I2C, I2S, UART, TWAI, and SDIO Host 3.0 buses, along with analog-to-digital conversion (ADC), pulse-width modulation (PWM), and other essential interfaces.

Despite the board's versatility and low cost, availability remains limited. "As ESP32-P4 processors are still available in limited quantities, only one board per order is allowed. Orders exceeding this limit will be put on hold and not processed," said Olimex founder Tsvetan Usunov.

The kit is priced at €16 (approximately \$17), plus shipping, and is currently available on the Olimex web store. True to Olimex's commitment to open hardware, all design files for the kit are published on GitHub under the CERN Open Hardware License Version 2 (Strongly Reciprocal).

"ESP32-P4-DevKit is our first development board with the new Espressif dual-core RISC-V processor [ESP32-P4]," said Usunov. "The ESP32-P4 has support in ESP-IDF 5.4, and we successfully built demos for Blink, iperf, MIPI-CSI, and SDmmc."

Optimised Ubuntu image launched for Qualcomm IoT platforms

Canonical and Qualcomm Technologies have announced the beta launch of the first-ever optimised Ubuntu image for Qualcomm IoT platforms. This program enables developers to access Ubuntu 22.04 LTS tailored for the Qualcomm RB3 Gen 2

Linux 6.12 released

Linux kernel 6.12, officially released as a Long Term Support (LTS) version, guarantees updates and support until 2026. This version introduces a host of improvements, spanning real-time computing, enhanced hardware support, networking capabilities, and file system upgrades, making it a significant milestone for Linux enthusiasts and enterprise users alike.

A standout feature in Linux 6.12 is the integration of the PREEMPT_RT real-time patchset, a decade-long development effort aimed at optimising task response times and enhancing overall system performance. This makes the kernel more suited for enterprise-grade real-time applications, with compatibility extended to ARM64, RISC-V, and X86/X86_64 architectures.

The kernel brings significant upgrades for hardware platforms like AMD, Intel and NVIDIA. For AMD, it adds RDNA 4 GPU support, OverDrive overclocking for SMU 14.x, and enhanced functionality for Instinct accelerators. For Intel, it includes support for Arrow Lake and Lunar Lake processors, improvements for hybrid P/E-core configurations, and early enablement for Panther Lake and Diamond Rapids Xeon processors. For NVIDIA, it implements virtual command queue support for SMMUv3.

New drivers include support for Microchip PIC64GX, NXP i.MX8ULP, Rockchip RK3568 CAN-FD controllers, Realtek automotive PCIe devices, and more. Updates also cover external hardware monitors and other critical peripherals.

The release introduces file-backed mount support for the EROFS file system, LoongArch KVM guest PMU support, enhanced Android guest capabilities, memory protection with ARM permission overlay, and integration with the latest Rust updates.

Linus Torvalds, the creator of Linux, confirmed the stable release, stating, "No strange surprises this last week, so we're sticking to the regular release schedule."